

Divide and Subtract Game

Difficulty: Medium | Accuracy: 42.52% | Submissions: 6K+ | Points: 4 | Average Time: 15m

Jon and Arya are playing a game. The rules of the game are as follows:

They start with a single number n . Both players make alternate moves, with Jon making the first move.

In each move, a player can perform exactly one of the following operations:

- Divide the current number by 2, 3, 4, or 5 and take the floor value of the result.
- Subtract 2, 3, 4, or 5 from the current number.

If after making a move the number becomes 1, the player who made that move automatically loses the game.

The game ends when the number becomes 0.

If a player cannot make a valid move, that player loses the game, determine the winner assuming both players play optimally.

Examples

Input: $n = 3$

Output: "Jon"

Explanation: Jon can subtract 3 from the initial number. The number becomes 0, and Arya has no valid move. Therefore, Jon wins.

Input: $n = 6$

Output: "Arya"

Explanation: Jon can divide 6 by 3, making the number 2. Arya can then subtract 2, making the number 0. Hence, Jon loses and Arya wins.

Constraints

$$2 \leq n \leq 10^5$$

Expected Complexities

Time Complexity: $O(n)$

Auxiliary Space: $O(1)$

Topic Tags

Game Theory

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